

System Architecture & Technical Methodology

ProfilerX addresses this by implementing an automated, multi-layered data ingestion and machine learning pipeline divided into four primary stages:

- **Automated OSINT Ingestion Engine:** The system initialises via a single seed input (e.g., a known alias, email prefix, or phone number). It triggers an asynchronous reconnaissance module modelled after advanced enumeration frameworks (such as Sherlock and WhatsMyName), querying over 300 web platforms simultaneously. Concurrently, an automated Google/Bing Dorking module scans indexed public directories, while metadata parsers extract EXIF data (GPS coordinates, camera serial numbers, and timestamps) from public media URLs.
- **NLP & Stylometric Feature Extraction:** Once raw profile data is scraped, it passes through a natural language processing (NLP) pipeline. Using Named Entity Recognition (NER) via spaCy and fine-tuned Transformer models, the system extracts unique entities (unlisted email addresses, geographic locations, and organisation names) from public bios and captions. To link accounts where handles differ entirely, a **Stylometry (Writeprint) Module** analyses behavioural text attributes, including vocabulary diversity, specific punctuation frequencies, emoji fingerprints, and common typing errors. Additionally, a **Temporal Activity Module** maps posting frequencies across a 24-hour cycle, normalising timestamps to identify localised active timezones.
- **Graph-Based Entity Resolution Engine:** The core intelligence layer aggregates all extracted features into a high-performance **Neo4j Graph Database**. Entities (profiles, email hashes, tracking IDs, phone fragments) are represented as nodes, while relationships (shared connections, interactions, similar posting times) represent edges. The engine uses a hybrid matching model to calculate a **Linkage Confidence Score (CS)**:
 - *Deterministic Matching:* Hard links based on exact cryptographic hashes of unique identifiers (e.g., matching SHA-256 email hashes or tracking parameters in shared URLs).
 - *Probabilistic Matching:* Soft links calculated using string distance metrics like the **Jaro-Winkler algorithm** for handle mutations (e.g., mapping @johndoe to @j0hn_doe), combined with network graph clustering algorithms (such as Louvain Community Detection) to detect shared high-density interaction circles.

3. Frontend Dashboard & Visualisation

The user interface is a web-based investigative terminal built using React.js and Next.js, integrating **Cytoscape.js** (or Vis.js) for real-world graph rendering. Investigators are presented with an interactive, node-based web of connected accounts. Clicking any node expands its associated metadata, showing temporal activity overlays, behavioural similarity indexes, and the calculated confidence score of the linkage. The platform includes a one-click "Case Export" feature that compiles an analytical, chain-of-custody compliant intelligence report detailing all discovered public footprints.

4. Impact & Expected Outcomes

ProfilerX transforms scattered, chaotic social media footprints into structured, actionable intelligence. By automating the correlation of public data points, it reduces the investigation timeline from weeks to seconds. The system operates entirely on publicly available data, ensuring full legal compliance with digital privacy standards while providing intelligence agencies, cyber investigators, and anti-fraud units with an unparalleled tool for suspect profiling and digital identity mapping.

References

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3)Linking Profiles Across Social Networks

<https://arxiv.org/abs/1605.05166>

4)Adversarial-Enhanced Hybrid Graph Network for User Identity Linkage (AHG-Net)

https://xuemensong.github.io/SIGIR_User.pdf